

# BOUNDARY LAYER CLIMATES

S E C O N D E D I T I O N

T. R. OKE



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# Boundary Layer Climates

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**BOUNDARY  
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CLIMATES**

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To my mother Kathleen  
and my wife Midge



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## Preface to first edition

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This book is designed to introduce students to the nature of the atmosphere near the ground. It is especially aimed at those whose curiosity about the atmosphere has been raised by an introductory course or general interest, but who are daunted by the technical nature of most micro- or biometeorological texts which assume a reasonably advanced ability in physics and mathematics. I believe that the frustration felt by such students can be alleviated by the more qualitative approach to the physics of the lower atmosphere used here. The book is based on the application of simple physical principles, and an exposition that is *explanatory* rather than *descriptive*. It is expected to appeal to two groups of students: first, those who want to know about the role of the atmosphere in environmental science and its applications in geography, agriculture, forestry, ecology, engineering and planning; second, those embarked on a more specialist course in small scale meteorology who wish to supplement their technical material with examples of atmospheric systems from many real world environments.

The book is organized in three sections. Part I—Atmospheric Systems—provides a simple scientific introduction to atmospheric processes operating in the planetary boundary layer. The idea of the flows and transformations of energy and mass through systems is developed. This supplies the basic concepts embodied in modern energy and water balance climatology and sets the framework for the subsequent analysis of the climates of a wide range of surface environments. Thus in Part II—Natural Atmospheric Environments—climates are presented by first considering the special mix of physical properties (radiative, thermal, moisture and aerodynamic) characterizing a particular system, then interpreting the effects they have upon the exchanges of energy, mass and momentum, before describing the resulting climatic qualities. Boundary layer climates are therefore viewed as the outcome of the unique way each surface responds to external forcing functions such as solar heating, precipitation and airflow. This provides a process-response (cause and effect) framework rather than the more traditional case study approach to understanding climates, and a rational basis for the analysis of the climatic impact of human activities in Part III—Man-Modified Atmospheric Environments.



The text attempts to be illustrative rather than comprehensive. The reader who requires further examples and case studies is recommended to consult Geiger's classic study *The Climate Near the Ground* (1965). For those with the appropriate scientific background who wish to extend their knowledge into the more technical fields of micro- and biometeorology Appendix A2 provides an introduction to some of the approaches used to evaluate quantitatively the flows of energy, mass and momentum. Throughout the book *Système International* (SI) units are used. Thus the basic units of length, mass and time are the metre, kilogram and second respectively. A fuller outline of SI units and their equivalents in other systems is given in Appendix A4. Mathematical equations have been kept to a minimum and the symbols used conform closely to those suggested by the World Meteorological Organization.

I am very grateful to the many colleagues who provided their views on early drafts of the manuscript (see Acknowledgements). Naturally I remain responsible for any remaining errors.

Finally I wish to thank my wife and children for affording me the time and the happy supportive environment necessary for writing.

*Vancouver*  
1977

T. R. O.

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## Preface to second edition

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Gratified by the generous reception accorded to the first edition of this book, and the helpful comments of many readers, I have been encouraged to provide this updated and slightly expanded edition.

The approach and objectives are retained. I have tried to keep the book pitched at about the same introductory level so that it continues to appeal across a range of disciplines.

The main changes include a re-casting of Chapters 1 and 2 with greater emphasis given to the layering of the atmosphere near the surface, the nature of thermals and convective cloud, atmospheric stability and the structure of the full planetary boundary layer. Chapter 4 includes new material on the climate of leaves, and also on vegetation roughness and evapotranspiration differences between crops and forests. Chapter 5 gives more attention than before to the effects of topography in modifying airflow. Chapter 8 has been up-dated to include developments in our understanding of airflow and the energy balance of cities and the effects of building geometry on the urban heat island. There is a completely new appendix dealing with radiation geometry and a section on the Combination Model for evapotranspiration. The reference tables have been expanded and there are almost 50 new figures.

*Vancouver*  
1987

T. R. O.



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# Symbols

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| <i>Symbol</i>                | <i>Quantity</i>                                                   | <i>SI Units</i>                             |
|------------------------------|-------------------------------------------------------------------|---------------------------------------------|
| <i>Roman capital letters</i> |                                                                   |                                             |
| <i>A</i>                     | horizontal moisture transport in the air per unit horizontal area | $\text{kg m}^{-2} \text{s}^{-1}$            |
| <i>A'</i>                    | lot area                                                          | $\text{m}^2$                                |
| <i>A*</i>                    | silhouette area                                                   | $\text{m}^2$                                |
| <i>B</i>                     | water intake by an animal                                         | $\text{kg}, \text{kg m}^{-2} \text{s}^{-1}$ |
| <i>C</i>                     | heat capacity of a substance                                      | $\text{J m}^{-3} \text{K}^{-1}$             |
| <i>D</i>                     | diffuse short-wave radiation                                      | $\text{W m}^{-2}$                           |
|                              | Dalton Number (Appendix A2)                                       |                                             |
| <i>E</i>                     | evapotranspiration                                                | $\text{mm}, \text{kg m}^{-2} \text{s}^{-1}$ |
| <i>ELR</i>                   | environmental lapse rate                                          | $\text{K m}^{-1}$                           |
| <i>F</i>                     | anthropogenic water release due to combustion (Chapter 8)         | $\text{mm}, \text{kg m}^{-2} \text{s}^{-1}$ |
| <i>F<sub>C</sub></i>         | carbon dioxide flux density                                       | $\text{kg m}^{-2} \text{s}^{-1}$            |
| <i>F<sub>P</sub></i>         | pollution flux density                                            | $\text{kg m}^{-2} \text{s}^{-1}$            |
| <i>H</i>                     | height                                                            | $\text{m}$                                  |
| <i>H<sub>s</sub></i>         | effective stack height (Chapter 9)                                | $\text{m}$                                  |
| <i>I</i>                     | pipd water supply per unit horizontal area (Chapter 8)            | $\text{mm}, \text{kg m}^{-2} \text{s}^{-1}$ |
|                              | radiant intensity (Appendix A1)                                   | $\text{W sr}^{-1}$                          |
| <i>I<sub>0</sub></i>         | solar constant                                                    | $\text{W m}^{-2}$                           |
| <i>J</i>                     | vertical flux of soil water                                       | $\text{kg m}^{-2} \text{s}^{-1}$            |
| <i>K*</i>                    | net short-wave radiation                                          | $\text{W m}^{-2}$                           |
| <i>K<sub>↓</sub></i>         | incoming short-wave radiation                                     | $\text{W m}^{-2}$                           |
| <i>K<sub>↑</sub></i>         | reflected short-wave radiation                                    | $\text{W m}^{-2}$                           |
| <i>K<sub>f</sub></i>         | hydraulic conductivity                                            | $\text{mm d}^{-1}$                          |
| <i>K<sub>C</sub></i>         | eddy diffusion coefficient for CO <sub>2</sub>                    | $\text{m}^2 \text{s}^{-1}$                  |

| <i>Symbol</i>    | <i>Quantity</i>                                                                          | <i>SI Units</i>                      |
|------------------|------------------------------------------------------------------------------------------|--------------------------------------|
| $K_{\text{Ex}}$  | extraterrestrial solar radiation                                                         | $\text{W m}^{-2}$                    |
| $K_{\text{H}}$   | eddy conductivity                                                                        | $\text{m}^2 \text{s}^{-1}$           |
| $K_{\text{M}}$   | eddy viscosity                                                                           | $\text{m}^2 \text{s}^{-1}$           |
| $K_{\text{P}}$   | eddy diffusion coefficient for pollution                                                 | $\text{m}^2 \text{s}^{-1}$           |
| $K_{\text{V}}$   | eddy diffusivity for water vapour                                                        | $\text{m}^2 \text{s}^{-1}$           |
| $L$              | latent heat, of fusion ( $L_f$ ), of vaporization ( $L_v$ ),<br>of sublimation ( $L_s$ ) | $\text{J kg}^{-1}$                   |
| $L^*$            | net long-wave radiation                                                                  | $\text{W m}^{-2}$                    |
| $L_{\downarrow}$ | incoming long-wave radiation from the atmosphere                                         | $\text{W m}^{-2}$                    |
| $L_{\uparrow}$   | outgoing long-wave radiation from a surface                                              | $\text{W m}^{-2}$                    |
| $M$              | flux density of water melt or freeze                                                     | $\text{mm, kg m}^{-2} \text{s}^{-1}$ |
| $N$              | diffuse irradiance                                                                       | $\text{W m}^{-2} \text{sr}^{-1}$     |
| $P$              | total atmospheric pressure                                                               | $\text{Pa}$                          |
|                  | rate of gross photosynthesis (Chapter 4)                                                 | $\text{kg m}^{-2} \text{s}^{-1}$     |
|                  | wave period (Chapter 2)                                                                  | $\text{s}$                           |
|                  | population of a settlement (Chapter 8)                                                   |                                      |
| $Q$              | heat energy                                                                              | $\text{J}$                           |
| $Q^*$            | net all-wave radiation flux density                                                      | $\text{W m}^{-2}$                    |
| $Q_{\downarrow}$ | total incoming short- and long-wave radiation                                            | $\text{W m}^{-2}$                    |
| $Q_{\uparrow}$   | total outgoing short- and long-wave radiation                                            | $\text{W m}^{-2}$                    |
| $Q_{\text{A}}$   | horizontal energy transport in the air per unit<br>horizontal area                       | $\text{W m}^{-2}$                    |
| $Q_{\text{E}}$   | turbulent latent heat flux density                                                       | $\text{W m}^{-2}$                    |
| $Q_{\text{F}}$   | anthropogenic heat flux density due to combustion                                        | $\text{W m}^{-2}$                    |
| $Q_{\text{G}}$   | sub-surface heat flux density                                                            | $\text{W m}^{-2}$                    |
| $Q_{\text{H}}$   | turbulent sensible heat flux density                                                     | $\text{W m}^{-2}$                    |
| $Q_{\text{M}}$   | metabolic heat production by animals<br>(Chapter 6)                                      | $\text{W m}^{-2}$                    |
| $Q_{\text{P}}$   | rate of energy storage in photosynthesis                                                 | $\text{W m}^{-2}$                    |
| $Q_{\text{R}}$   | rate of heat supply by rainfall                                                          | $\text{W m}^{-2}$                    |
| $R$              | rate of $\text{CO}_2$ respiration by plants                                              | $\text{kg m}^{-2} \text{s}^{-1}$     |
| $Ri$             | Richardson's Number                                                                      |                                      |
| $S$              | direct-beam short-wave radiation                                                         | $\text{W m}^{-2}$                    |
|                  | soil moisture content or water storage                                                   |                                      |
| $T$              | temperature                                                                              | $\text{K, } (^{\circ}\text{C})$      |
| $T_{\text{b}}$   | animal body (or core) temperature                                                        | $\text{K, } (^{\circ}\text{C})$      |
| $U$              | urinary water loss by an animal                                                          | $\text{kg, kg m}^{-2} \text{s}^{-1}$ |
| $V$              | voltage                                                                                  | $\text{V}$                           |
| $W$              | width                                                                                    | $\text{m}$                           |
| $X$              | length                                                                                   | $\text{m}$                           |
| $Z$              | solar zenith angle                                                                       |                                      |

*Symbol Quantity*

*SI Units*

*Roman small letters*

|          |                                                                             |                                                         |
|----------|-----------------------------------------------------------------------------|---------------------------------------------------------|
| $a$      | extinction coefficient                                                      | $\text{m}^{-1}$                                         |
| $c$      | specific heat of a substance                                                | $\text{J kg}^{-1} \text{K}^{-1}$                        |
| $c_p$    | specific heat of air at constant pressure                                   | $\text{J kg}^{-1} \text{K}^{-1}$                        |
| $d$      | zero plane displacement                                                     | $\text{m}$                                              |
| $e$      | vapour pressure; saturation value ( $e^*$ )<br>base of Napierian logarithms | $\text{Pa}$                                             |
| $f$      | moisture infiltration                                                       | $\text{mm, kg m}^2 \text{s}^{-1}$                       |
| $g$      | acceleration due to gravity                                                 | $\text{m s}^{-2}$                                       |
| $h$      | height of an object (e.g. crop or shelterbelt)<br>hour angle                | $\text{m}$                                              |
| $h^*$    | depth of the surface mixed layer                                            | $\text{m}$                                              |
| $h_s$    | stack height                                                                | $\text{m}$                                              |
| $k$      | thermal conductivity<br>von Karman's constant                               | $\text{W m}^{-1} \text{K}^{-1}$<br>$\text{m}$           |
| $m$      | optical air mass number                                                     |                                                         |
| $p$      | precipitation                                                               | $\text{mm}$                                             |
| $r$      | resistance to flow<br>radius                                                | $\text{s m}^{-1}$                                       |
| $r_a$    | aerodynamic resistance                                                      | $\text{s m}^{-1}$                                       |
| $r_b$    | laminar boundary layer resistance                                           | $\text{s m}^{-1}$                                       |
| $r_c$    | canopy (or surface) resistance                                              | $\text{s m}^{-1}$                                       |
| $r_m$    | mesophyll resistance                                                        | $\text{s m}^{-1}$                                       |
| $r_{st}$ | stomatal resistance                                                         | $\text{s m}^{-1}$                                       |
| $s$      | slope of the saturation vapour versus temperature curve                     | $\text{Pa K}^{-1},$<br>$\text{kg m}^{-3} \text{K}^{-1}$ |
| $t$      | time                                                                        | $\text{s}$                                              |
| $u$      | horizontal wind speed<br>optical depth of water vapour                      | $\text{m s}^{-1}$<br>$\text{mm}$                        |
| $u_*$    | friction velocity                                                           | $\text{m s}^{-1}$                                       |
| $v_{dd}$ | vapour density deficit                                                      | $\text{kg m}^{-3}$                                      |
| $v_{pd}$ | vapour pressure deficit                                                     | $\text{Pa}$                                             |
| $w$      | vertical wind speed                                                         | $\text{m s}^{-1}$                                       |
| $x$      | horizontal (along-wind) distance                                            | $\text{m}$                                              |
| $y$      | horizontal (across-wind) distance                                           | $\text{m}$                                              |
| $z$      | vertical distance                                                           | $\text{m}$                                              |
| $z_0$    | roughness length                                                            | $\text{m}$                                              |

| <i>Symbol</i>                    | <i>Quantity</i>                                                                                         | <i>SI Units</i>                                 |
|----------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <i>Greek capital letters</i>     |                                                                                                         |                                                 |
| $\Gamma$                         | dry adiabatic lapse rate                                                                                | $K m^{-1}$                                      |
| $\Delta$                         | finite difference approximation (i.e. difference or net change in a quantity)                           |                                                 |
| $\Delta A$                       | net moisture advection; rate per unit volume (per unit horizontal area)                                 | $kg; kg m^{-3} s^{-1}$<br>$(kg m^{-2} s^{-1})$  |
| $\Delta P$                       | net rate of photosynthesis, rate of net $CO_2$ assimilation                                             | $kg m^{-2} s^{-1}$                              |
| $\Delta Q_A$                     | net energy (sensible and latent) advection; rate per unit volume (per unit horizontal area)             | $J; W m^{-3}$<br>$(W m^{-2})$                   |
| $\Delta Q_M$                     | net latent heat storage change due to melting/freezing; rate per unit volume (per unit horizontal area) | $J; W m^{-3}$<br>$(W m^{-2})$                   |
| $\Delta Q_P$                     | net energy storage due to photosynthesis; rate per unit volume (per unit horizontal area)               | $J; W m^{-3}$<br>$(W m^{-2})$                   |
| $\Delta Q_S$                     | net energy storage; rate per unit volume (per unit horizontal area)                                     | $J; W m^{-3}$<br>$(W m^{-2})$                   |
| $\Delta S$                       | net moisture storage; rate per unit volume (per unit horizontal area)                                   | $kg; kg m^{-3} s^{-1};$<br>$(kg m^{-2} s^{-1})$ |
| $\Delta h$                       | height of plume rise                                                                                    | $m$                                             |
| $\Delta r$                       | net run off                                                                                             | $mm, kg m^{-2} s^{-1}$                          |
| $\Theta$                         | zenith angle                                                                                            |                                                 |
| $\Phi_C, \Phi_H, \Phi_M, \Phi_V$ | dimensionless stability functions for carbon dioxide, heat, momentum and water vapour respectively.     |                                                 |
| $X$                              | rate of pollution emission                                                                              | $kg m^{-2} s^{-1},$<br>$m^3 s^{-1}$             |
| $\Psi$                           | soil moisture potential                                                                                 | $Pa$                                            |
|                                  | transmissivity                                                                                          |                                                 |
| $\Omega$                         | solar azimuth angle                                                                                     |                                                 |

*Greek small letters*

|          |                                                |
|----------|------------------------------------------------|
| $\alpha$ | surface albedo                                 |
| $\beta$  | Bowen's ratio ( $Q_H/Q_E$ )<br>elevation angle |

*Symbol Quantity*

|                                |                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------|
| $\gamma$                       | psychrometric constant                                                                                            |
| $\gamma^*$                     | apparent psychrometric constant                                                                                   |
| $\delta$                       | solar declination angle                                                                                           |
| $\varepsilon$                  | surface emissivity                                                                                                |
| $\zeta$                        | absorptivity                                                                                                      |
| $\theta$                       | potential temperature                                                                                             |
| $\kappa$                       | thermal diffusivity of a substance                                                                                |
| $\kappa_H, \kappa_M, \kappa_V$ | molecular diffusion coefficients for heat, momentum (viscosity) and water vapour in air                           |
| $\lambda$                      | wavelength                                                                                                        |
| $\mu$                          | thermal admittance                                                                                                |
| $\rho$                         | density of a substance: of air $\rho_a$ ; of CO <sub>2</sub> $\rho_c$ ; of water vapour $\rho_v$ ; water $\rho_w$ |
| $\rho_v^*$                     | saturation water vapour density                                                                                   |
| $\sigma$                       | Stefan-Boltzmann constant                                                                                         |
| $\sigma_y, \sigma_z$           | standard deviations of horizontal and vertical pollutant distribution                                             |
| $\tau$                         | momentum flux per unit surface area                                                                               |
| $\phi$                         | heat of assimilation of carbon<br>latitude                                                                        |
| $\chi$                         | concentration of an air pollutant                                                                                 |
| $\psi$                         | view factor                                                                                                       |
| $\omega$                       | solid angle                                                                                                       |

*SI Units*

|                                                 |
|-------------------------------------------------|
| $\text{Pa K}^{-1}$ ;                            |
| $\text{kg m}^{-3} \text{K}^{-1}$                |
| $\text{Pa K}^{-1}$ ;                            |
| $\text{kg m}^{-3} \text{K}^{-1}$                |
| K                                               |
| $\text{m}^2 \text{s}^{-1}$                      |
| $\text{m}^2 \text{s}^{-1}$                      |
| m                                               |
| $\text{J m}^{-2} \text{s}^{-1/2} \text{K}^{-1}$ |
| $\text{kg m}^{-3}$                              |
| $\text{kg m}^{-3}$                              |
| $\text{W m}^{-2} \text{K}^{-4}$                 |
| m                                               |
| Pa                                              |
| $\text{J kg}^{-1}$                              |
| $\text{kg m}^{-3}$ , ppm                        |
| sr                                              |

*Common subscripts*

|      |                                    |
|------|------------------------------------|
| (A)  | Atmosphere                         |
| (E)  | Earth                              |
| a    | air                                |
| (b)  | bottom (abaxial) surface of a leaf |
| c    | cloud<br>canopy                    |
| f    | forest floor; fusion               |
| h    | horizontal                         |
| i    | normal incidence                   |
| l    | liquid                             |
| long | long-wave                          |
| max  | maximum                            |
| min  | minimum                            |



*Symbol Quantity*

*SI Units*

|           |                                 |
|-----------|---------------------------------|
| o         | surface                         |
| (o)       | cloudless                       |
| r         | rural                           |
| s         | soil; sublimation               |
| short     | short-wave                      |
| sky       | sky                             |
| (t)       | top (adaxial) surface of a leaf |
| u         | urban                           |
| v         | vapour                          |
| $\lambda$ | wavelength                      |

*Superscripts*

|          |                                   |
|----------|-----------------------------------|
| $\wedge$ | sloping surface                   |
| —        | time average                      |
| '        | instantaneous deviation from mean |
| $\sim$   | spatial average                   |

# **Part I**

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## **Atmospheric systems**

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# Energy and mass exchanges

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## 1 ATMOSPHERIC SCALES

The Atmosphere is characterized by phenomena whose space and time scales cover a very wide range. The space scales of these features are determined by their typical size or wavelength, and the time scales by their typical lifetime or period. Figure 1.1 is an attempt to place various atmospheric phenomena (mainly associated with motion) within a grid of their probable space and time limits. The features range from small-scale turbulence (tiny swirling eddies with very short life spans) in the lower left-hand corner, all the way up to jet streams (giant waves of wind encircling the whole Earth) in the upper right-hand corner.

In reality none of these phenomena is discrete but part of a continuum, therefore it is not surprising that attempts to divide atmospheric phenomena into distinct classes have resulted in disagreement with regard to the scale limits. Most classification schemes use the characteristic horizontal distance scale as the sole criterion. A reasonable consensus of these schemes gives the following scales and their limits (see the top of Figure 1.1):

Micro-scale  $10^{-2}$  to  $10^3$  m  
Local scale  $10^2$  to  $5 \times 10^4$  m  
Meso-scale  $10^4$  to  $2 \times 10^5$  m  
Macro-scale  $10^5$  to  $10^8$  m

In these terms this book is mainly restricted to atmospheric features whose horizontal extent falls within the micro- and local scale categories. A fuller description of its scope is given by also including characteristic vertical distance, and time scales.

This book is concerned with the interaction between the Atmosphere and the Earth's surface. The influence of the surface is effectively limited to the lowest 10 km of the Atmosphere in a layer called the *troposphere* (Figure 1.2). (*Note*—terms introduced for the first time are italicized and the meaning

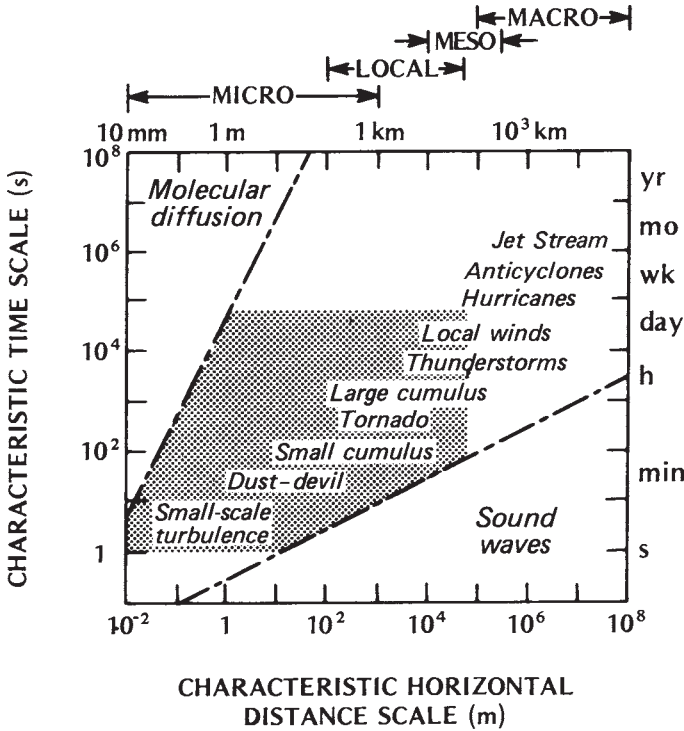


Figure 1.1 Time and space scales of various atmospheric phenomena. The shaded area represents the characteristic domain of boundary layer features (modified after Smagorinsky, 1974).

of those not fully explained in the text is given in Appendix A5.) Over time periods of about one day this influence is restricted to a very much shallower zone known as the *planetary or atmospheric boundary layer*, hereinafter referred to simply as the *boundary layer*. This layer is particularly characterized by well developed mixing (turbulence) generated by frictional drag as the Atmosphere moves across the rough and rigid surface of the Earth, and by the ‘bubbling-up’ of air parcels from the heated surface. The boundary layer receives much of its heat and all of its water through this process of turbulence.

The height of the boundary layer (i.e. the depth of surface-related influence) is not constant with time, it depends upon the strength of the surface-generated mixing. By day, when the Earth’s surface is heated by the Sun, there is an upward transfer of heat into the cooler Atmosphere. This vigorous thermal mixing (convection) enables the boundary layer depth to extend to about 1 to 2 km. Conversely by night, when the Earth’s surface cools more rapidly than the Atmosphere, there is a downward transfer of

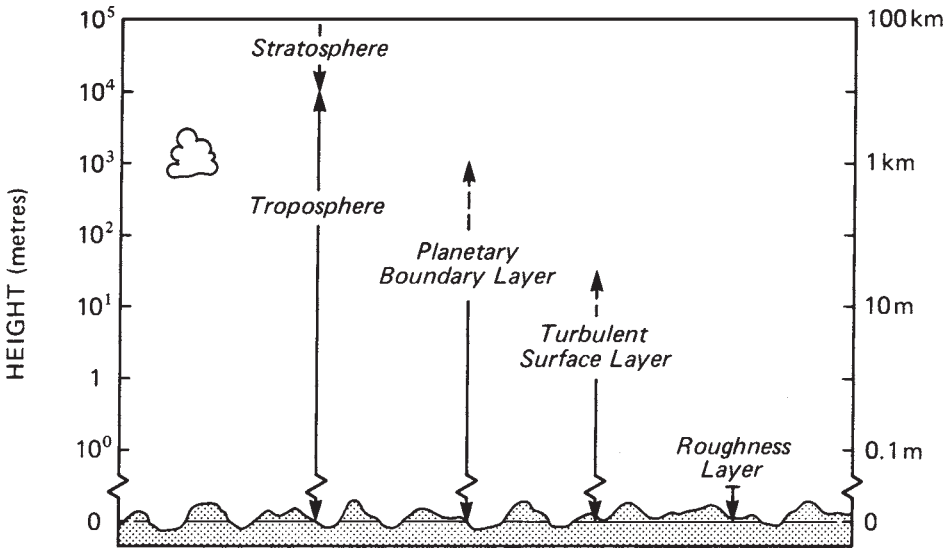


Figure 1.2 The vertical structure of the atmosphere.

heat. This tends to suppress mixing and the boundary layer depth may shrink to less than 100 m. Thus in the simple case we envisage a layer of influence which waxes and wanes in a rhythmic fashion in response to the daily solar cycle.

Naturally this ideal picture can be considerably disrupted by large-scale weather systems whose wind and cloud patterns are not tied to surface features, or to the daily heating cycle. For our purposes the characteristic horizontal distance scale for the boundary layer can be related to the distance air can travel during a heating or cooling portion of the daily cycle. Since significant thermal differences only develop if the wind speed is light (say less than  $5 \text{ ms}^{-1}$ ) this places an upper horizontal scale limit of about 50 to 100 km. With strong winds mixing is so effective that small-scale surface differences are obliterated. Then, except for the dynamic interaction between airflow and the terrain, the boundary layer characteristics are dominated by tropospheric controls. In summary the upper scale limits of boundary layer phenomena (and the subject matter of this book) are vertical and horizontal distances of  $\sim 1 \text{ km}$  and  $\sim 50 \text{ km}$  respectively, and a time period of  $\sim 1 \text{ day}$ .

The *turbulent surface layer* (Figure 1.2) is characterized by intense small-scale turbulence generated by the surface roughness and convection; by day it may extend to a height of about 50 m, but at night when the boundary layer shrinks it may be only a few metres in depth. Despite its variability in the short term (e.g. seconds) the surface layer is horizontally homogeneous when viewed over longer periods (greater than 10 min). Beneath the surface

## 6 Boundary Layer Climates

layer are two others that are controlled by surface features, and whose depths are dependent upon the dimensions of the surface roughness elements. The first is the *roughness layer* which extends above the tops of the elements to at least 1 to 3 times their height or spacing. In this zone the flow is highly irregular being strongly affected by the nature of the individual roughness features (e.g. blades of grass, trees, buildings, etc.). The second is the *laminar boundary layer* which is in direct contact with the surface(s). It is the non-turbulent layer, at most a few millimetres thick, that adheres to all surfaces and establishes a buffer between the surface and the more freely diffusive environment above. The dimensions of the laminar boundary layer define the lower vertical size scale for this book.

The lower horizontal scale limit is dictated by the dimensions of relevant surface units and since the smallest climates covered are those of insects and leaves, this limit is of the order of  $10^{-2}$  to  $10^{-3}$  m. It is difficult to set an objective lower cut-off for the time scale. An arbitrary period of approximately 1 s is suggested.

The shaded area in Figure 1.1 gives some notion of the space and time bounds to boundary layer climates as discussed in this book (except that it requires a third co-ordinate to show the vertical space scale). Two aberrations from this format should be noted. First, it should be pointed out that precipitation and violent weather events (such as tornadoes), which might be classed as boundary layer phenomena, have been omitted. The former, although deriving their initial impetus near the surface, owe their internal dynamics to condensation which often occurs at the top or above the boundary layer. The latter are dominated by weather dynamics occurring at much larger scales than outlined above. Second, the boundary layer treated herein includes the uppermost layer of the underlying material (soil, water, snow, etc.) extending to a depth where diurnal exchanges of water and heat become negligible.

## 2 A SYSTEMS VIEW OF ENERGY AND MASS EXCHANGES AND BALANCES

The classical climatology practised in the first half of the twentieth century was almost entirely concerned with the distribution of the principal climatological parameters (e.g. air temperature and humidity) in time and space. While this information conveys a useful impression of the state of the atmosphere at a location it does little to explain how this came about. Such parameters are really only indirect measures of more fundamental quantities. Air temperature and humidity are really a gauge of the thermal energy and water status of the atmosphere respectively, and these are tied to the fundamental energy and water cycles of the *Earth-Atmosphere system*. Study of these cycles, involving the processes by which energy and mass

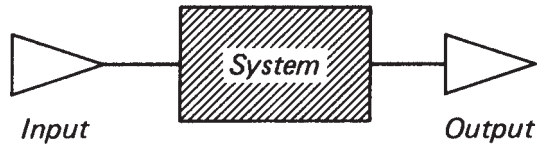


Figure 1.3 Energy flow through a system.

are transferred, converted and stored, forms the basis of modern physical climatology.

The relationship between energy flow and the climate can be illustrated in the following simple manner. The First Law of Thermodynamics (conservation of energy) states that energy can be neither created nor destroyed, only converted from one form to another. This means that for a simple system such as that in Figure 1.3, two possibilities exist. Firstly:

$$\text{Energy Input} = \text{Energy Output} \quad (1.1)$$

in this case there is no change in the net energy status of the system through which the energy has passed. It should however be realized that this does not mean that the system has no energy, merely that *no change* has taken place. Neither does it mean that the Output energy is necessarily in the same *form* as it was when it entered. Energy of importance to climatology exists in the Earth-Atmosphere system in four different forms (*radiant, thermal, kinetic* and *potential*) and is continually being transformed from one to another. Hence, for example, the Input energy might be entirely radiant but the Output might be a mixture of all four forms. Equally the Input and Output *modes* of energy transport may be very different. The exchange of energy within the Earth-Atmosphere system is possible in three modes (*conduction, convection* and *radiation* (see Section 3 for explanation)).

The second possibility in Figure 1.3 is:

$$\text{Energy Input} = \text{Energy Output} + \text{Energy Storage Change}$$

For most natural systems the equality,  $\text{Input} = \text{Output}$ , is only valid if values are integrated over a long period of time (e.g. a year). Over shorter periods the energy balance of the system differs significantly from equality. The difference is accounted for by energy accumulation or depletion in the system's energy store. (The energy storage term may have a positive or negative sign. By convention a positive storage indicates the addition of energy.) In climatic terms, for example, if energy is being accumulated in a soil-atmosphere system it probably means an increase in soil and/or air temperature.

Hence we see the link between process (energy flow) and response (temperature change). The whole relationship is referred to as a *process-response system*, which in essence describes the connection between cause



## 8 Boundary Layer Climates

and effect. The degree of detailed understanding of the system depends on how well the internal workings of the 'box' in Figure 1.3 are known. Inside the box the energy is likely to be channelled into different subsystems, and converted into different combinations of energy forms and modes of transport. Some will lead to energy storage change and others to energy output from the system. This partitioning is not haphazard, it is a function of the system's physical properties. In the case of energy these properties include its ability to absorb, transmit, reflect and emit radiation, its ability to conduct and convect heat, and its capacity to store energy.

In the analogous case of water flow in a soil-atmosphere system the mass of water is conserved at all times but it may be found in three different *states* (vapour, liquid and solid); be transported in a number of modes (including convection, precipitation, percolation, and runoff); and its accumulation or depletion in stores is measured as changes of water content (atmospheric humidity, soil moisture or the water equivalent of a snow or ice mass). Similar analogues can be extended to the mass balances of other substances cycled through systems as a result of natural or human (anthropogenic) activities including sulphur, carbon, nitrogen, and particulates. In the case of atmospheric systems the accumulation of these substances beyond certain levels constitutes atmospheric pollution. This occurs when the natural cycling of substances is upset by human activities. For example, in urban areas, if the emission (input) of these materials exceeds the physical capability of the local atmospheric system to flush itself (output) in a short period of time, the result is an increase in the local concentration of that substance (i.e. increased storage). Therefore in the most general form we may write the following energy or mass balance equation for a system:

$$\text{Input-Output-Storage Change}=0 \quad (1.2)$$

There are two fundamental cycles of importance in understanding atmospheric systems. These are the cycles of solar energy (heat), and water (mass). The remainder of this chapter is concerned with a description of the workings of these two cycles. This is followed in Chapter 2 by an explanation of the way these exchange processes and balances are linked to the vertical distributions of such climatological elements as temperature, humidity and wind speed in the boundary layer.

### 3 ENERGY BALANCES

#### (a) Radiation characteristics

Radiation is a form of energy due to the rapid oscillations of electromagnetic fields. It is transferred by *photons*, or bundles of energy that have properties

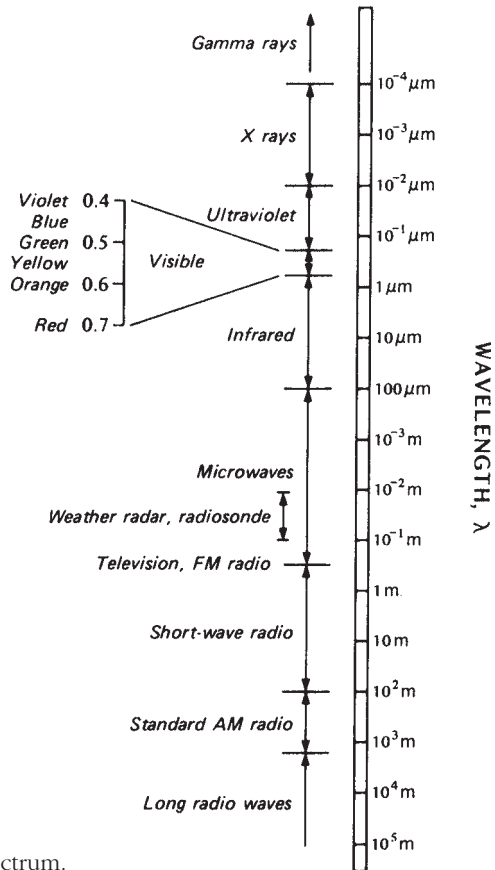


Figure 1.4 The electromagnetic spectrum.

similar to both particles and waves. The oscillations may be considered as travelling waves characterized by their wavelength  $\lambda$  (distance between successive wavecrests). In most atmospheric applications we are concerned with wavelengths in the approximate range  $0.1$  to  $100 \mu\text{m}$  ( $1 \mu\text{m} = 10^{-6} \text{m}$ ), representing only a very small portion of the total electromagnetic spectrum (Figure 1.4). The visible portion of the spectrum, to which the human eye is sensitive, is an even smaller fraction extending from  $0.36 \mu\text{m}$  (violet) to  $0.75 \mu\text{m}$  (red). Radiation is able to travel in a vacuum, and all radiation moves at the speed of light ( $3 \times 10^8 \text{ms}^{-1}$ ), and in a straight path. The wavelength is uniquely related to the photon energy so that it is possible to calculate the photon energy flux at any given wavelength or waveband (see Appendix A4f, p. 399).

All bodies possessing energy (i.e. whose temperatures are above absolute zero,  $0 \text{K} = -273.2^\circ\text{C}$ , see p. 395) emit radiation. If a body at a given temperature emits the maximum possible amount of radiation per unit of its surface area

in unit time then it is called a *black body* or *full radiator*. Such a body has a surface *emissivity* ( $\epsilon$ ) equal to unity.

Less efficient radiators have emissivities between zero and unity. The relation between the amount of radiation emitted by a black body, and the wavelength of that radiation at a given temperature is given by Planck's Law. In graphical form this law shows the spectral distribution of radiation from a full radiator to be a characteristic curve (Figure 1.5). The shape consists of a single peak of emission at one wavelength ( $\lambda_{\max}$ ), and a tailing-off at increasingly higher wavelengths. The form is so characteristic that in Figure 1.5 the same Planck curve on different scales, describes the emission spectra from full radiators at 300 and 6000 K. However, the total amount of radiation given out and its spectral composition are very different for the two cases.

The total energy emitted by each body in Figure 1.5 is proportional to the area under the curve (including the tail at longer wavelengths that has been truncated). This is the basis of the Stefan-Boltzmann Law:

$$\text{Energy emitted} = \sigma T_0^4 \quad (1.3)$$

where,  $\sigma$ —Stefan-Boltzmann proportionality constant= $5.67 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$ , and  $T_0$ —surface temperature of the body (K). In the typical range of temperatures encountered in the E-A system ( $-15$  to  $45^\circ\text{C}$ ) a change of 1 K in  $T_0$  of a full radiator results in a change of the emitted radiation of between 4 and 7  $\text{Wm}^{-2}$  (see Appendix A3, p. 394). If the body is not a full radiator, equation 1.3 can be re-written to include the value of the surface emissivity:

$$\text{Energy emitted} = \epsilon \sigma T_0^4. \quad (1.4)$$

Note that the energy emission given by these equations is the *radiant flux* (rate of flow of radiation) ( $\text{Js}^{-1}=\text{W}$ ) from unit area ( $\text{m}^2$ ) of a plane surface into the overlying hemisphere. The flux per unit area of a quantity is termed its *flux density* ( $\text{Wm}^{-2}$ ). Further, *irradiance* is the radiant flux density incident on a surface whereas *emittance* is the radiant flux density emitted by a surface.

The effect of temperature change on the wavelength composition of the emitted radiation is embodied in Wien's Displacement Law. It states that a rise in the temperature of a body not only increases the total radiant output, but also increases the proportion of shorter wavelengths of which it is composed. Thus as the temperature of a full radiator increases, the Planck curve is progressively shifted to the left, and the wavelength of peak emission ( $\lambda_{\max}$ ) moves with it so that:

$$\lambda_{\max} = 2.88 \times 10^{-3} / T_0 \quad (1.5)$$

with  $\lambda_{\max}$  in metres and  $T_0$  on the Kelvin scale.

The temperatures in Figure 1.5 were chosen because they approximately represent the average surface temperatures of the Sun and the E-A system,

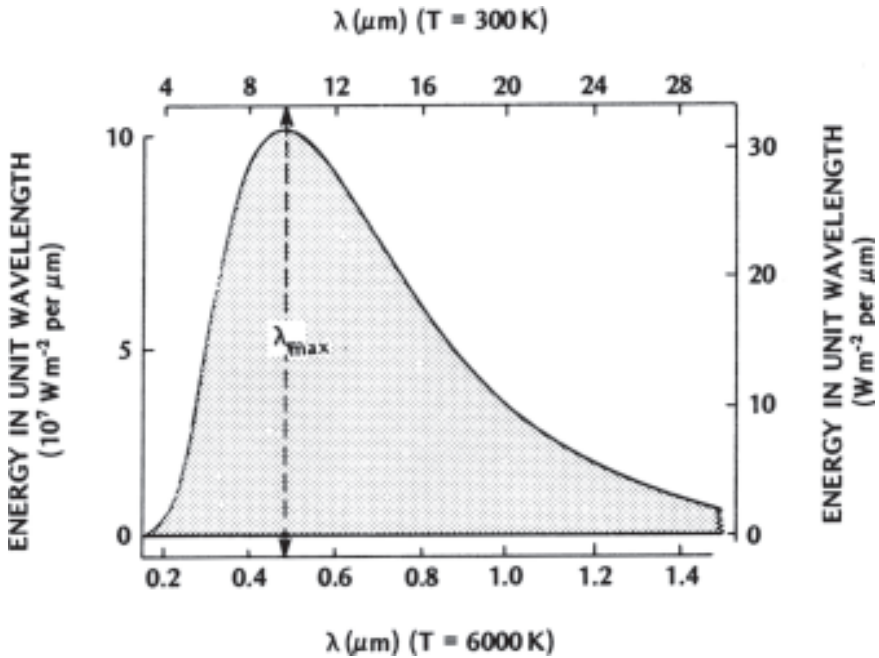


Figure 1.5 Spectral distribution of radiant energy from a full radiator at a temperature of (a) 6000 K, left-hand vertical and lower horizontal axis, and (b) 300 K, right-hand vertical and upper horizontal axis.  $\lambda_{\text{max}}$  is the wavelength at which energy output per unit wavelength is maximal (after Monteith, 1973).

and thus illustrate the nature of the radiation each emits. Obviously the E-A system emits smaller total amounts of energy than the Sun, but also the wavelength composition is very different. From equation 1.5 it can be seen that the Sun's peak wavelength is about  $0.48\text{ }\mu\text{m}$  (in the middle of the visible spectrum), whereas for the E-A system it is about  $10\text{ }\mu\text{m}$ . Typical wavelengths for radiation from the Sun extend from  $0.15\text{ }\mu\text{m}$  (ultra-violet) to about  $3.0\text{ }\mu\text{m}$  (near infra-red), whereas E-A system radiant wavelengths extend from  $3.0\text{ }\mu\text{m}$  to about  $100\text{ }\mu\text{m}$ , well into the infra-red. In fact the difference between the two radiation regimes is conveniently distinct; about 99% of the total energy emitted by the two planets lies within these limits. On this basis atmospheric scientists have designated the radiation observed in the range  $0.15\text{--}3.0\text{ }\mu\text{m}$  to be *short-wave* or *solar radiation*, and that in the range  $3.0\text{--}100\text{ }\mu\text{m}$  to be *long-wave* radiation.

Radiation of wavelength  $\lambda$  incident upon a substance must either be transmitted through it or be reflected from its surface, or be absorbed. This is a statement of the conservation of energy. By expressing the proportions transmitted, reflected and absorbed as ratios of the incident energy, we

## 12 Boundary Layer Climates

Table 1.1 Radiative properties of natural materials.

| Surface                       | Remarks                                  | Albedo<br>$\alpha$     | Emissivity<br>$\epsilon$ |
|-------------------------------|------------------------------------------|------------------------|--------------------------|
| Soils                         | Dark, wet<br>Light, dry                  | 0.05–<br>0.40          | 0.98–<br>0.90            |
| Desert                        |                                          | 0.20–0.45              | 0.84–0.91                |
| Grass                         | Long (1.0 m)<br>Short (0.02 m)           | 0.16–<br>0.26          | 0.90–<br>0.95            |
| Agricultural crops,<br>tundra |                                          | 0.18–0.25              | 0.90–0.99                |
| Orchards                      |                                          | 0.15–0.20              |                          |
| Forests                       |                                          |                        |                          |
| Deciduous                     | Bare<br>Leaved                           | 0.15–<br>0.20          | 0.97–<br>0.98            |
| Coniferous                    |                                          | 0.05–0.15              | 0.97–0.99                |
| Water                         | Small zenith angle<br>Large zenith angle | 0.03–0.10<br>0.10–1.00 | 0.92–0.97<br>0.92–0.97   |
| Snow                          | Old<br>Fresh                             | 0.40–<br>0.95          | 0.82–<br>0.99            |
| Ice                           | Sea<br>Glacier                           | 0.30–0.45<br>0.20–0.40 | 0.92–0.97                |

Sources: Sellers (1965), List (1966), Paterson (1969) and Monteith (1973).

define the transmissivity ( $\psi_\lambda$ ), the reflectivity ( $\alpha_\lambda$ ) and the absorptivity ( $\zeta_\lambda$ ) and it follows that:

$$\psi_\lambda + \alpha_\lambda + \zeta_\lambda = 1 \quad (1.6)$$

These are radiative properties of the substance (expressed as dimensionless numbers between zero and unity). Strictly equation 1.6 is only valid for the case of a single wavelength. In practice it is usually acceptable for fairly wide wavebands (e.g. for solar radiation  $\alpha$  is referred to as the surface *albedo*—see Table 1.1 for typical values). It is however essential that each property refers to the same incident radiation.

It can also be shown that for the same radiation:

$$\zeta_\lambda = \epsilon_\lambda \quad (1.7)$$

This is Kirchhoff's Law, which holds that at the same temperature and wavelength good absorbers are good emitters. It follows that for a full radiator  $\zeta_\lambda = \epsilon_\lambda = 1$ , and from equation (1.6)  $\alpha_\lambda = \psi_\lambda = 0$ . Further, for an

opaque ( $\psi_\lambda=0$ ) non-black body there is some reflection which is given by:

$$\alpha_\lambda = 1 - \zeta_\lambda = 1 - \epsilon_\lambda \quad (1.8)$$

In boundary layer climatology these relations are very helpful in long-wave exchange considerations (i.e. between bodies at typical E-A system temperatures). Although they are theoretically valid for short-wave exchange, their use does not arise because no E-A system bodies emit short wavelength radiation. Typical values of long-wave surface emissivity ( $\epsilon$ ) for natural surfaces are given in Table 1.1. It is readily apparent that most of these surfaces are close to being full radiators ( $\epsilon$  typically greater than 0.90), so that reflection of long-wave radiation is small (i.e. from equation 1.8  $\alpha_{\text{long}}$  is generally less than 0.10).

During its passage through the Atmosphere the solar beam encounters clouds and other atmospheric constituents including water vapour, salt crystals, dust particles and various gases. Each of these constituents has its own set of radiative properties with respect to the incident short-wave radiation, thus part of the beam is reflected (scattered), a part is absorbed and the rest is transmitted to the surface. The ratio of the extraterrestrial input to these amounts defines the atmospheric reflectivity, absorptivity and transmissivity ( $\alpha_a$ ,  $\zeta_a$  and  $\psi_a$ ).

The nature and amount of absorption depends on the absorption spectra of the atmospheric gases (Figure 1.6) and of cloud and other aerosols. Figure 1.6 demonstrates that the Atmosphere is not a very good absorber of short-wave radiation (0.15-3.0  $\mu\text{m}$ ). Ozone ( $\text{O}_3$ ) is very effective at filtering out ultra-violet radiation at wavelengths less than 0.3  $\mu\text{m}$ , and water vapour becomes increasingly important at greater than 0.8  $\mu\text{m}$ , but in the intervening band where the intensity of solar radiation is greatest (i.e. near  $\lambda_{\text{max}}$  in Figure 1.5) the Atmosphere is relatively transparent. Even the absorption by liquid water drops in cloud is relatively small.

The portion of the incoming solar radiation that is reflected and scattered, together with that multiply-reflected between the surface and the atmosphere (back-scattered), gives *diffuse* short-wave radiation (D). As an approximation we may consider this radiant receipt to arrive from all parts of the sky hemisphere, although in cloudless conditions it is greater from the area of the sky around the solar disc and near the horizon. Clouds are very effective at diffusing short-wave radiation.

Finally, the portion of the incoming solar radiation that arrives at the Earth's surface, without being absorbed or diffused, is called the *direct-beam* short-wave radiation (S). Since it can be approximated as a parallel beam, the irradiance of an exposed surface depends on its orientation to the beam such that:

$$S = S_0 \cos \Theta \quad (1.9)$$

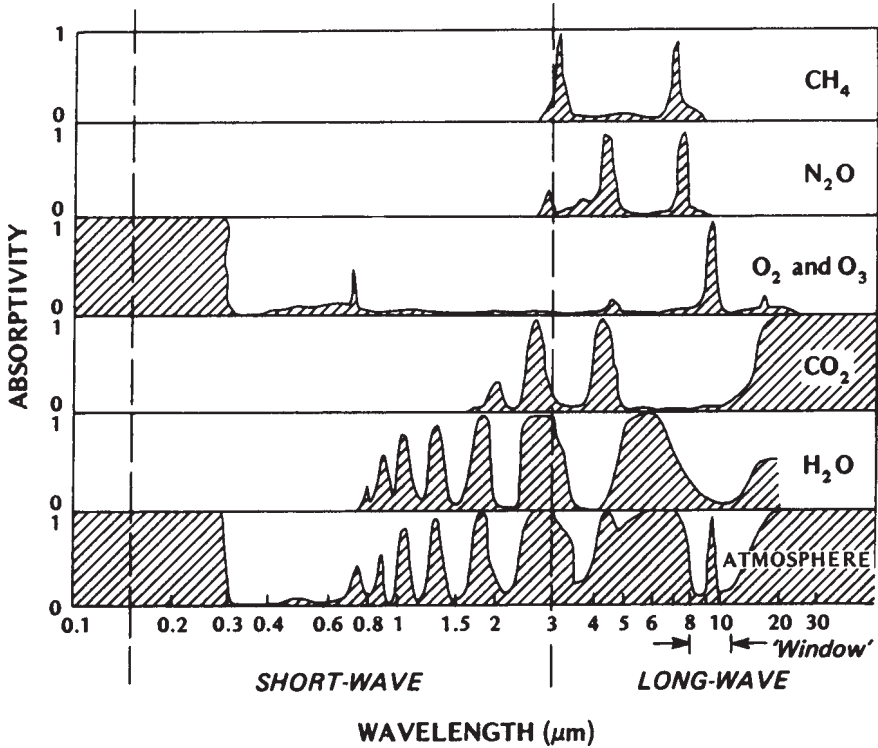


Figure 1.6 Absorption at various wavelengths by constituents of the Atmosphere, and by the Atmosphere as a whole (after Fleagle and Businger, 1963).

where  $S$  is the flux density of the beam radiation at the surface,  $S_i$  is the flux density normal to the beam and  $\Theta$  is the angle between the beam and the normal to the surface. This corollary of Lambert's Law (Appendix A1, p. 350), called the *cosine law of illumination*, is illustrated in Figure 1.7a, which shows that the greater the angle to the surface the larger is the area over which it is spread and hence the less the irradiance. (Note—in order to generalize equation 1.9 it is necessary to know the surface geometry and the azimuth and zenith angles of the Sun, see Appendix A1.)

The total short-wave radiation received at the surface ( $K\downarrow$ ) is simply:

$$K\downarrow = S + D \quad (1.10)$$

as illustrated in Figure 1.7b.

The process of long-wave radiative exchange in the Atmosphere is very complex. At all levels the Atmosphere absorbs long-wave radiation arriving from below (emitted from the surface and lower layers of air and cloud) and from above (higher layers of air and cloud). The absorption depends upon the long-wave absorptivities of the constituents present. In general

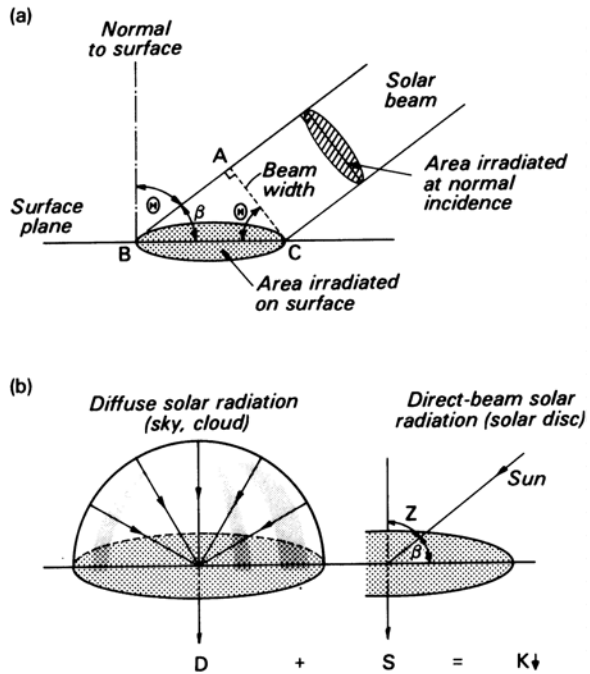


Figure 1.7 (a) Illustration of the areas irradiated by a circular beam on planes placed normal to, and at an angle  $\Theta$  to, the beam. The radiant energy flux ( $\text{Js}^{-1}$ ) is spread over unit area ( $=\pi(0.5AC)^2$ ) at normal incidence but over a larger area ( $=\pi(0.5BC)^2$ ) on the surface. The flux density ( $\text{Wm}^{-2}$ ) on the surface ( $S$ ) is less than that at normal incidence ( $S_i$ ) by the ratio  $AC/BC=\cos\Theta$  or  $\sin\beta$ . Therefore,  $S=S_i\cos\Theta$  and when  $\Theta=0^\circ\cos\Theta=1$  and  $S=S_i$ . For a horizontal surface  $\Theta=Z$  the zenith angle of the Sun. (b) The components of incoming solar radiation at the Earth's surface (modified after Monteith, 1973).

the Atmosphere is a relatively good absorber in the long-wave band (3 to 100  $\mu\text{m}$ ). As shown in Figure 1.6, this is particularly due to the absorptivities of water vapour ( $\text{H}_2\text{O}$ ), carbon dioxide ( $\text{CO}_2$ ) and ozone ( $\text{O}_3$ ). Of these, water vapour is by far the most important. If liquid water is present, as cloud droplets, the absorptivity is even greater. There is, however, one important gap in a cloudless Atmosphere's absorption spectrum for long-wave radiation. Except for a narrow band of ozone absorption (9.6 to 9.8  $\mu\text{m}$ ), the Atmosphere is open to the transmission of radiation in the 8 to 11  $\mu\text{m}$  band. This gap is called the *atmospheric 'window'*. It is through this 'window' that most of the E-A system longwave loss to Space occurs.



However, even this 'window' can be partially closed by clouds or atmospheric pollutants.

At all levels the atmosphere emits long-wave radiation consistent with its temperature ( $T_a$ ) and emissivity ( $\epsilon_a$ ) in accord with equation 1.4. We might also note that from Kirchhoff's Law since  $\zeta_\lambda = \epsilon_\lambda$ , just as the Atmosphere does not absorb in the 'window' region, it also will not emit those wavelengths. The atmospheric emission is directed both upwards and downwards. The processes of absorption and re-emission take place on a continuous basis throughout the Atmosphere, but quantitatively they are most important in the lowest layers where the concentrations of water vapour and carbon dioxide are greatest. The net portion which emerges from the top of the Atmosphere is lost to Space, and that which arrives at the Earth's surface is sometimes referred to as *counter-radiation* ( $L\downarrow$ ) because it counteracts the outgoing long-wave radiation from the surface ( $L\uparrow$ ). As with the diffuse solar radiation it is usually acceptable to assume that  $L\downarrow$  arrives equally from all parts of the sky hemisphere. In reality it is greatest from areas near the horizon and least from the zenith (directly overhead the point of concern).

### *Conduction*

Thermal *conduction* is the process whereby heat is transmitted within a substance by the collision of rapidly moving molecules. It is usually an effective mode of transfer in solids, less so in liquids and least important in gases. In general pure molecular conduction is negligible in atmospheric applications, except within the very thin laminar boundary layer (p. 37). On the other hand it is very important to the transport of heat beneath the surface. The conduction of heat is dependent on the thermal properties of the substrate. Theoretical considerations are covered in Chapter 2.

### *Convection*

The process of *convection* involves the vertical interchange of air masses and can only occur in liquids and gases. In the Atmosphere the parcels of air (or *eddies*) transport energy and mass from one location to another. The eddies may be set into turbulent motion by free or forced convection. *Free convection* is due to the parcel of air being at a different density than the surrounding fluid. If for example a parcel is warmer than its surroundings, it will be at a lower density and will tend to rise. Conversely, if it is cooler it will be denser and tend to sink. The motion of water in a heated kettle is free convection, and a similar 'bubbling-up' of air parcels occurs when the Earth's surface is strongly heated by solar radiation. If the state of the Atmosphere is conducive to free convection it is said to be *unstable*, and if it inhibits such motion it is *stable* (see p. 51). The

atmosphere near the Earth's surface may also be physically thrown into motion when it flows over obstacles. This is *forced*, or *mechanical, convection* and depends upon the roughness of the surface and the speed of the horizontal flow. Often free and forced convection co-exist giving *mixed convection*.

If the addition or subtraction of energy to a body is sensed as a rise or fall in its temperature then it is referred to as *sensible heat*. On the other hand, to enable a substance to change from liquid at a given temperature to vapour *at the same temperature*, requires the addition of heat. This heat which is not sensed as a temperature change is called *latent heat*. It is locked up within the substance and is available for release should the substance revert to its former state. Energy is taken up to move in the direction of a higher energy state (e.g. solid to liquid, or liquid to vapour) and released in moving in the opposite direction.

Convection transports heat to and from the Atmosphere in both its sensible and latent forms. Sensible heat is carried from a warmer surface into the cooler air above by turbulent eddies and is released when it mixes with the environmental air. The reverse transport occurs when the air is warmer than the surface. Latent heat transport is tied up with that of water vapour. The latent heat imparted to a parcel of moist air in the evaporation of water at the surface is liberated to warm the air when the water vapour condenses into cloud. Convection provides the means of transport and mixing.

This process is also responsible for the exchanges of carbon dioxide and pollutants between the surface and the Atmosphere and also for the extraction of momentum from the mean flow. The essential principles governing convective (turbulent) transport in the boundary layer are outlined in Chapter 2.

## **(b) Energy balance of the total Earth-Atmosphere system**

In this section we will use the annual energy balance of the Earth-Atmosphere system to illustrate the linkages between the various energy exchanges and the concept of energy balance. In so doing we will establish both the magnitude of the driving force for the E-A hydrologic cycle (see Section 4(b)) and the energetic context within which all E-A system climates (macro-, meso-, local and micro-) operate.

A schematic depiction of the annual energy balance of the E-A system is given in Figure 1.8. It recognizes the Earth, the Atmosphere and Space as separate sub-systems and places magnitudes on the energy exchanges between them. The E-A system is a *closed* one (i.e. it is closed to the import or export of mass, but it does allow exchange of energy with the exterior (Space)). The constant stream of radiant energy emitted by the Sun is the sole input to the system. The magnitude of this input, known as the *solar*

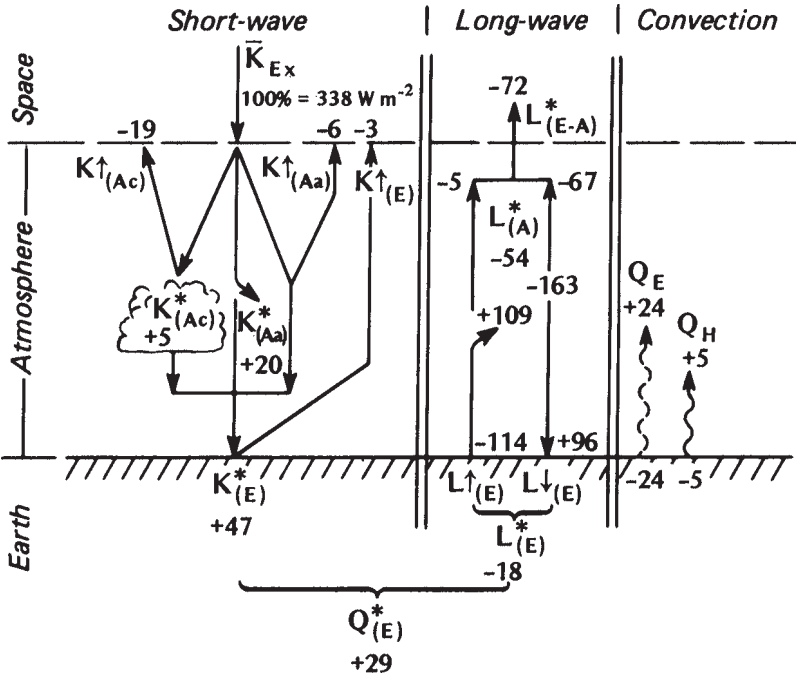


Figure 1.8 Schematic diagram of the average annual solar energy cascade of the Earth-Atmosphere system. Values are expressed as percentages of the average annual extra-terrestrial solar radiation ( $=342 \text{ W m}^{-2}$ ) (data from Rotty and Mitchell, 1974).

constant ( $I_0$ ), is  $1367 \text{ W m}^{-2}$  (Wehrli, 1985). It is the value observed outside the Atmosphere on a plane surface placed normal to the solar beam. This represents a practical upper limit for short-wave radiation in the E-A system because the depletion of the Atmosphere is not included and the definition includes the ideal orientation of the receiving surface (i.e. in equation 1.9  $\cos\Theta=1$ ). When averaged over the top of the Atmosphere for one year, the spatial mean input ( $\bar{K}_{\text{Ex}}$ ) is exactly  $I_0/4= 342 \text{ W m}^{-2}$  ( $29.5 \text{ MJ m}^{-2}\text{day}^{-1}$ ). In Figure 1.8 all fluxes are represented as percentages of this value. Over the period of a year exactly the same amount of energy must be lost from the E-A system to Space. If this were not so the system would experience a net energy gain or loss, resulting in a net storage change and a rise or fall of the average E-A system temperature (i.e. a climatic shift). Equally, if the sub-systems were not in balance the system would be in disequilibrium. By tracing the energy pathways we will see how balance is achieved.

In the Atmosphere clouds reflect about 19% of  $\bar{K}_{\text{Ex}}$  back to Space ( $K_{\text{Ac}}^{\uparrow}$ ) and absorb about 5% ( $K_{\text{Ac}}^*$ ). Atmospheric constituents scatter and reflect about 6% to Space ( $K_{\text{Aa}}^{\uparrow}$ ) and absorb about 20% ( $K_{\text{Aa}}^*$ ). The remainder of

the original beam is transmitted to the Earth's surface where approximately 3% is reflected to Space ( $K_{(E)}$ ) and the remaining 47% is absorbed ( $K_{(E)}^*$ ). Thus, in summary, the solar radiation input is disposed of in the following manner:

$$\begin{array}{ccccccc} \bar{K}_{Ex} = K_{\uparrow(Ac)} + K_{\uparrow(Aa)} + K_{\uparrow(Ac)}^* + K_{\uparrow(Aa)}^* + K_{\uparrow(E)} + K_{(E)}^* \\ 100 = 19 + 6 + 5 + 20 + 3 + 47 \\ \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & \text{Earth} & & \text{Earth} \\ \text{Atmospheric} & & \text{Atmospheric} & & \text{reflection} & & \text{absorption} \\ \text{reflection} & & \text{absorption} & & & & \end{array}$$

Three basic features of the short-wave radiation portion of the balance emerge. First, 28% of the E-A input is reflected to Space and does not participate further in the E-A system energy balance. Secondly, only 25% of the input is absorbed by the Atmosphere. Thus, as noted earlier, the Atmosphere is semi-transparent to short-wave radiation and consequently is not greatly heated by it. Thirdly, almost one-half (47%) of the input is absorbed at the Earth's surface. This considerable amount of energy is converted from radiation into thermal energy which warms the surface.

The Earth's surface emits long-wave radiation in accord with equation 1.4. Given that most natural surfaces have emissivities close to unity (Table 1.1), and that the Earth's mean annual temperature is approximately 288 K, this results in an upward emission ( $L_{\uparrow(E)}$ ) of 114% of  $\bar{K}_{Ex}$ . This apparent anomaly is possible because the Atmosphere blocks the loss of  $L_{\uparrow(E)}$  and forces the surface temperature above the value it would otherwise have with no Atmosphere. In fact only 5% is lost directly to Space, the remaining 109% being absorbed by the Atmosphere. It should also be noted that the Earth emits long-wave over its entire surface area, but only receives short-wave over the sunlit hemisphere. The Atmosphere emits long-wave radiation to Space (67%) and to the Earth's surface,  $L_{\downarrow(E)}$  (96%), amounting to a total output of 163%.

Let us summarize the radiation budgets of the E-A system and the Earth and Atmosphere sub-systems. The whole E-A system is in radiative equilibrium because the solar input (100%) is matched exactly by the sum of the short-wave scattering and reflection (19+6+3=28%) and the long-wave emission from Earth and its Atmosphere (5+67=72%).

The sub-systems are not in equilibrium. The Earth's surface receives a net input of short-wave radiation ( $K_{(E)}^*$ ) equivalent to 47% of  $\bar{K}_{Ex}$ , but experiences a net loss of long-wave radiation ( $L_{(E)}^*$ ) of 18% (because it emits 114% to the Atmosphere but receives 96% in return). Thus the net all-wave radiation budget for the Earth ( $Q_{(E)}^*$ ) is positive and represents 29% (47-18) of the original extra-terrestrial input. In the case of the Atmosphere it gains 25% as  $K_{(A)}^*$  due to absorption by clouds and atmospheric constituents

(see equation 1.7), but loses 54% as  $L^*_{(A)}$  (because although it absorbs 109% from the surface, it emits 163% to Space and back to the surface). Thus the net all-wave radiation budget of the atmosphere ( $Q^*_{(A)}$ ) is -29% (25–54).

Therefore the Earth has an annual radiant energy surplus of 29% and the Atmosphere has an annual radiant energy deficit of the same amount. This does not mean there is a balance. Considering their respective physical and thermal properties, this situation would result in the Earth warming up at the rate of approximately  $250^\circ\text{C day}^{-1}$  and the Atmosphere cooling at approximately  $1^\circ\text{C day}^{-1}$ . Such heating and cooling rates are not observed because convection transports energy equivalent to the radiative surplus of the Earth into the Atmosphere thereby offsetting its deficit; 5% of the exchange is as sensible heat ( $Q_H$ ) and 24% is as latent heat ( $Q_E$ ). For completeness we should note that conduction does not appear in Figure 1.8 because over the annual time period net sub-surface storage is zero.

### **(c) Diurnal energy balance at an ‘ideal’ site**

In Section 1 it was noted that boundary layer climates respond to processes operating on time scales of less than one day. This section outlines the most important general features of the diurnal energy regime at a given site. This is best accomplished by considering the case of an ‘ideal’ site. Such a location presents the minimum complication being horizontal, homogeneous and extensive. These constraints ensure that surface/atmosphere fluxes are spatially uniform and confined to the vertical direction. To minimize fluctuations in the time domain only cloudless conditions are considered initially, so that the solar input is a smooth wave. The surface is a flat, moist, bare soil (or short grass) located in the mid-latitudes in the warm season. In Part II of the book these constraints are removed.

Figure 1.9 shows the diurnal variation of the important radiation budget components at a site meeting our ‘ideal’ criteria, and the accompanying table summarizes the daily energy totals. The pattern of incoming short-wave radiation ( $K\downarrow$ ) is controlled by the azimuth ( $\Omega$ ) and zenith ( $Z$ ) angles of the Sun relative to the horizon, with a maximum at local solar noon. (Definitions of these angles and methods to calculate them for any latitude, time of year and hour are given in Appendix A1 together with information concerning the path of the Sun across the sky and the effect of horizon obstruction.)

In the middle of a cloudless day the proportion of  $K\downarrow$  arriving as diffuse radiation is anywhere from 10 to 25% depending on the amount of water vapour haze; in smoggy urban and industrial areas it will be even greater. Early and late in the day the diffuse proportion also increases due to the greater path length of the Sun through the Atmosphere (see Figure 1.12).

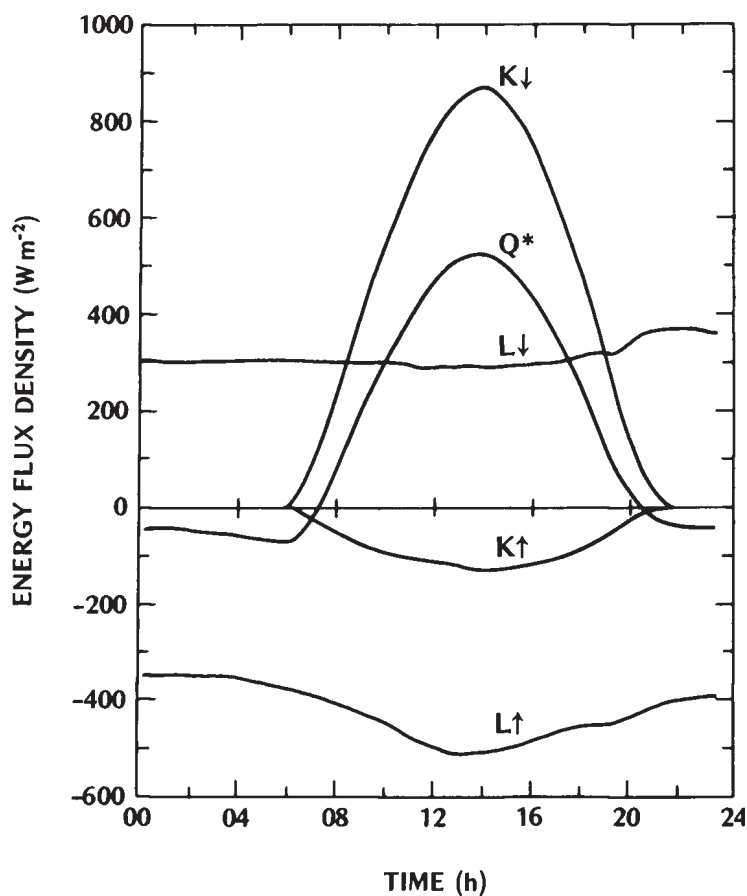


Figure 1.9 Radiation budget components for 30 July 1971, at Matador, Saskatchewan (50°N) over a 0.2 m stand of native grass. Cloudless skies in the morning, increasing cloud in the later afternoon and evening (after Ripley and Redmann, 1976). (Note—In the text no signs have been given to individual radiation fluxes, only to net fluxes ( $K^*$ ,  $L^*$  and  $Q^*$ ). However, in figures such as this radiative inputs to the surface ( $K\downarrow$ ,  $L\downarrow$ ) have been plotted as positive, and outputs ( $K\uparrow$ ,  $L\uparrow$ ) as negative to aid interpretation.) The following table gives the radiation totals for the day ( $\text{MJ m}^{-2}\text{day}^{-1}$ ).

|                  |      |               |      |
|------------------|------|---------------|------|
| $K\downarrow$    | 27.3 | $L\downarrow$ | 27.5 |
| $K\uparrow$      | 4.5  | $L\uparrow$   | 36.8 |
| $K^*$            | 22.7 | $L^*$         | -9.3 |
| $\alpha^\dagger$ | 0.16 | $Q^*$         | 13.4 |

$^\dagger$  Dimensionless

In a relatively clean atmosphere approximately 50% of  $K\downarrow$  is in the visible portion of the electromagnetic spectrum.

The short-wave radiation reflected from the surface ( $K\uparrow$ ) depends on the amount of incident radiation ( $K\downarrow$ ) and the surface albedo ( $\alpha$ , see Table 1.1, p. 12):

$$K\uparrow = K\downarrow(\alpha) \quad (1.11)$$

Although  $\alpha$  is not a perfect constant through a day (e.g. see p. 86 and Figures 3.12 and 4.14) to a first approximation, it is reasonable to expect  $K\uparrow$  to be a reduced mirror-image of  $K\downarrow$  (Figure 1.9). Given that the surface is opaque to short-wave radiation (i.e.  $\psi_{\text{short}}=0$ ), the portion of  $K\downarrow$  that is not reflected is absorbed, so the net short-wave radiation ( $K^*$ ) is:

$$\begin{aligned} K^* &= K\downarrow - K\uparrow \\ &= K\downarrow(1-\alpha) \end{aligned} \quad (1.12)$$

Therefore in our example, since  $\alpha=0.16$ ,  $K\downarrow$  would describe a curve with positive values of about  $0.84K\downarrow$ , in Figure 1.9.

The incoming long-wave radiation emitted by the atmosphere ( $L\downarrow$ ) in the absence of cloud depends upon the bulk atmospheric temperature and emissivity (which itself depends on the distributions of temperature, water vapour and carbon dioxide) in accord with the Stefan-Boltzmann Law (equation 1.4). Neither of these properties fluctuates rapidly and hence  $L\downarrow$  is almost constant through the day (see Figure 1.9; note the increase of  $L\downarrow$  between 18 and 24 h is discussed on p. 26).

The outgoing long-wave radiation from the surface ( $L\uparrow$ ) is similarly governed by its temperature and emissivity. If the surface is a full radiator ( $\epsilon_0=1$ ) the output is given by equation 1.3, but if  $\epsilon_0$  is less than unity:

$$L\uparrow = \epsilon_0 \sigma T_0^4 + (1 - \epsilon_0) L\downarrow \quad (1.13)$$

The second term on the right accounts for the amount of  $L\downarrow$  that is reflected (see equation 1.8). For most surfaces the adjustment is small. Because the values of  $T_0$  and  $\epsilon_0$  are greater than their atmospheric counterparts, and because  $T_0$  varies considerably through the day, the value of  $L\uparrow$  is both greater in magnitude and more variable than  $L\downarrow$ .

The difference between the two long-wave fluxes is the surface net long-wave radiation budget ( $L^*$ ):

$$L^* = L\downarrow - L\uparrow \quad (1.14)$$

The value of  $L^*$  (not plotted in Figure 1.9) is usually negative, and relatively small (75 to 125  $\text{Wm}^{-2}$ ) if the surface and air temperatures are not significantly different. If the surface is considerably warmer than the air (e.g. as in Figure 7.3, p. 233)  $L^*$  may be much larger. The diurnal course of  $L^*$  is usually in phase with  $L\uparrow$ .

The net all-wave radiation ( $Q^*$ ) is the most important energy exchange because for most systems it represents the limit to the available energy source or sink. The daytime surface budget is the sum of the individual short- and long-wave streams:

$$\begin{aligned} Q^* &= K\downarrow - K\uparrow + L\downarrow - L\uparrow \\ &= K^* + L^* \end{aligned} \quad (1.15)$$

and, at night solar radiation is absent so that:

$$\begin{aligned} Q^* &= L\downarrow - L\uparrow \\ &= L^* \end{aligned} \quad (1.16)$$

Thus the typical diurnal course of  $Q^*$  (Figure 1.9) involves a daytime surface radiant surplus when the net short-wave gain exceeds the net long-wave loss; and a nocturnal surface deficit when the net long-wave loss is unopposed by solar input. At a given location the terms  $K\downarrow$ , and  $L\downarrow$ , are unlikely to show significant spatial variability because they are governed by large-scale atmospheric, or Earth-Sun geometric relationships. On the other hand  $K\uparrow$  and  $L\uparrow$  are governed by sensitive site specific factors (i.e.  $K\uparrow$  by  $\alpha$ ;  $L\uparrow$  by  $T_0$  and  $\epsilon_0$ ). Thus it is these terms which govern the differences in radiation budget ( $Q^*$ ) between surfaces in the same local region. In conclusion it should be noted that the range of  $Q^*$  values over different surfaces is damped somewhat by a built-in negative feedback mechanism. The range of natural surface  $\epsilon_0$  values is small (Table 1.1) and hence differences in  $Q^*$  effectively depend upon the values of  $\alpha$  and  $T_0$ . A surface with a low albedo will absorb well, but unless it possesses channels for rapid heat dissipation this will result in a high surface temperature. Thus the large  $K^*$  gain will be matched, at least in part, by a large  $L^*$  loss.

Appendix A2 provides examples of the currently utilized methods (measurement and calculation) for the determination of the surface radiation balance fluxes, and the relevant surface radiative properties.

The net all-wave radiation flux is not only the end result of the radiation budget but also the basic input to the surface energy balance. Figure 1.10a shows the typical diurnal variation of the components of the surface energy balance at an 'ideal' site, and its table summarizes the daily energy totals. At any given time it can be seen that any surface radiative imbalance is accounted for by a combination of convective exchange to or from the atmosphere, either as sensible ( $Q_H$ ) or latent heat ( $Q_E$ ), and conduction to or from the underlying soil ( $Q_G$ ). Thus the surface energy balance is:

$$Q^* = Q_H + Q_E + Q_G \quad (1.17)$$

The sign convention employed in Figure 1.10a (and throughout the remainder of the book) is that non-radiative fluxes directed away from a



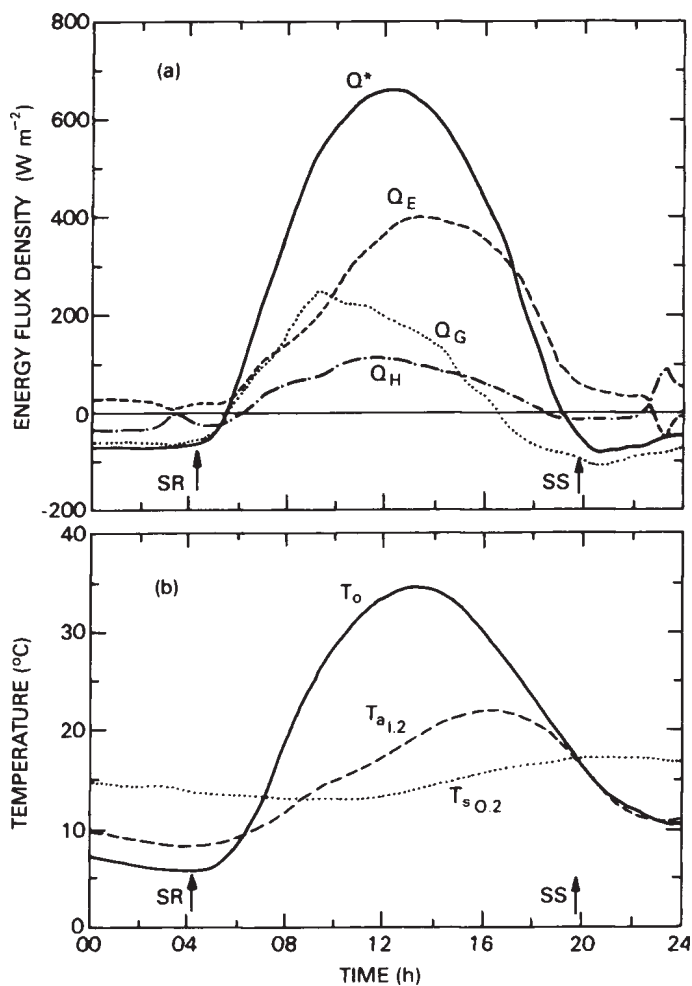


Figure 1.10 (a) Energy balance components for 30 May 1978 with cloudless skies at Agassiz, B.C. ( $49^{\circ}\text{N}$ ) for a moist, bare soil, and (b) temperatures at the surface, in the air at a height of 1.2 m and in the soil at a depth of 0.2 m (after Novak and Black, 1985). The following table gives the energy totals for the day ( $\text{MJm}^{-2}\text{day}^{-1}$ ).

| Energy balance |      | Derived terms  |      |
|----------------|------|----------------|------|
| $Q^*$          | 18.0 | $\beta$        | 0.17 |
| $Q_H$          | 2.3  | $Q_E/Q^*$      | 0.75 |
| $Q_E$          | 13.4 | $E(\text{mm})$ | 5.45 |
| $Q_G$          | 2.3  |                |      |

surface (or system) are positive. Thus the terms on the right-hand side of equation 1.17 are positive when they represent losses of heat for the surface (or system), and negative when they are gains. On the left-hand side  $Q^*$  is positive as a gain and negative when a loss. When both sides of the equation are positive it describes how the available radiative surplus is partitioned into sub-surface and atmospheric energy sinks; and this is usually the situation by day. When both sides are negative the equation states how the surface radiative deficit is partitioned between heat gain from available sub-surface and atmospheric sources; and this is the normal nocturnal situation. The flux of momentum is an exception to this convention (see Chapter 2).

The exact partitioning of the radiative surplus or deficit (between  $Q_H$ ,  $Q_E$  and  $Q_G$ ) is governed by the nature of the surface, and the relative abilities of the soil and atmosphere to transport heat. The particular apportionment arrived at by a surface is probably the most important determinant of its microclimate.

The diurnal course of  $Q^*$  in Figure 1.10a is very similar to that of Figure 1.9. *Under the given conditions* the daytime  $Q^*$  is dissipated as  $Q_E$ ,  $Q_G$  and  $Q_H$  in descending order of importance. The dominant role of  $Q_E$  is a result of the free availability of soil moisture for evaporation at this irrigated site. If water became more restricted we might expect the role of  $Q_E$  to drop, and of  $Q_H$  to rise. No matter which dominates it is clear that convection is the principal means of daytime heat transport away from the interface.

At night on the other hand the situation is reversed. The nocturnal  $Q^*$  loss is most effectively replenished by conduction upwards from the soil, and the convective contribution is least effective from  $Q_E$ . The essential difference between the two convective situations is due to the fact that by day free convection is enhanced, but by night it is damped by the atmospheric temperature stratification (p. 51). The size of  $Q_G$  is not greatly different between day and night. In fact, although  $Q_G$  is a significant energy source, or sink, on an hourly basis, when integrated over the full day its net effect is not large (see the table accompanying Figure 1.10). In summer the daytime storage slightly exceeds the nocturnal output and the soil gradually warms. The reverse is true in winter.

Figure 1.11 provides a convenient schematic summary of the terms involved in the surface radiation and energy budgets of an 'ideal' site.

Before concluding this section a few remarks should be added concerning the effects of cloud, and non-uniform surface properties, upon the 'ideal' situation described above. Clouds exert a major influence on the exchanges of short- and long-wave radiation. The surface receipt of  $K?$  is reduced because of cloud absorption and the reflection from cloud tops, and with partly cloudy skies short-term variability becomes great. For example, in

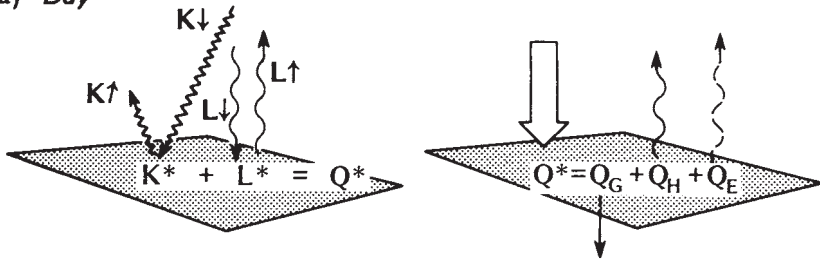
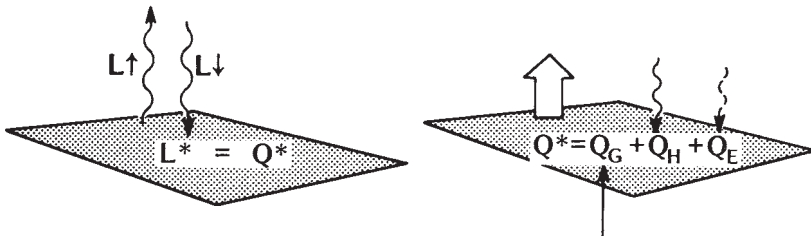
**(a) Day****(b) Night**

Figure 1.11 Schematic summary of the fluxes involved in the radiation budget and energy balance of an 'ideal' site, (a) by day and (b) at night.

the afternoon of the day shown in Figure 1.12 the very rapid fluctuations in  $K\downarrow$  are due to the almost complete elimination of  $S$  by cloud. At the same time there are relative increases in  $D$ . Notice also the short-term bursts of  $K\downarrow$  that *exceed* the cloudless values by about 5–10%. These occur when the recording station is in receipt of direct-beam and diffuse sky radiation as normal plus reflection from the sides of isolated cumulus clouds in the vicinity. With a complete overcast of low, thick cloud  $K\downarrow$  can be reduced to 10% of the cloudless value and beam irradiation is completely eliminated. Contrary to the cloudless case,  $D$  is greater from the zenith than from near the horizon.

The surface long-wave budget is profoundly affected because clouds are almost black bodies (p. 15) and thus absorb and emit very efficiently. Clouds therefore absorb much of  $L\uparrow$  from the surface and re-emit it back so that  $L\downarrow$  is enhanced, and  $L^*$  reduced. The arrival of cloud explains the abrupt increase of  $L\downarrow$  noted in Figure 1.9 after 18 h. The cloud emission depends on the cloud-base temperature, therefore the effect of Stratus (low, relatively warm) is much greater than Altus or Cirrus (high, cold) cloud. The net result of cloud is to damp the diurnal surface radiation budget variation, and serves to reduce the diurnal temperature range. This explains why cloudy weather is associated with comparatively uniform temperatures because daytime solar heating and night-time long-wave cooling are both reduced.

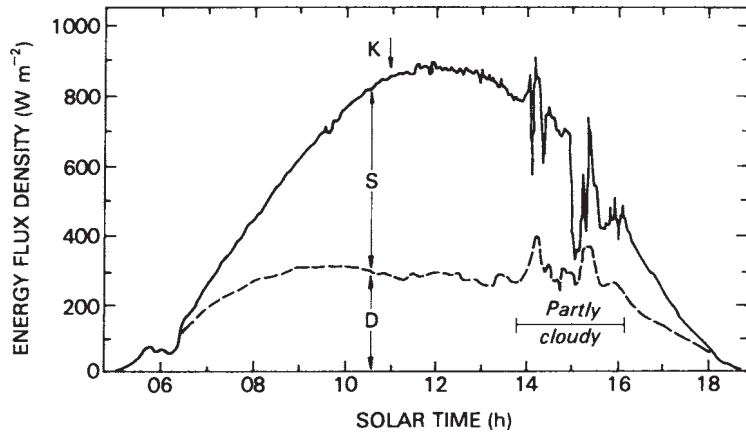


Figure 1.12 Variation of incoming solar radiation ( $K$ ) on a very hazy day (10 August 1975) in central Illinois ( $39^{\circ}\text{N}$ ), including the distinction between direct-beam ( $S$ ) and diffuse radiation ( $D$ ) (modified after Wesely, 1982).

If the site is not sufficiently extensive it is possible that heat exchange can occur between it and upwind surfaces possessing different energy partitioning. For example, if our grass site was downwind of a hot dry soil surface it is possible that horizontal airflow could carry air with greater sensible heat content (and therefore higher temperature) across the grass. This would alter the atmospheric conditions and give rise to an adjustment in the surface energy fluxes. In the example used it would tend to suppress the local surface value of  $Q$  and to augment  $Q$  (this is explained more fully in Chapter 5). The net horizontal convective heat transport (both sensible and latent) is called *advection* ( $\Delta Q$ ). Unless specifically noted it may be assumed that the surface climates outlined in Part II are advection-free. In reality this is rarely completely true.

#### (d) Atmospheric motion

Horizontal temperature variations in the E-A system give rise to horizontal pressure differences, which result in motion (winds). In this way thermal energy from the solar energy cycle is converted into the kinetic energy of wind systems. The energy then participates in the *kinetic energy* cascade involving the transfer of energy to increasingly small scales of motion by turbulence. This is the sequence depicted in Figure 1.1. Kinetic energy enters the cascade at a size-scale governed by the forces generating the motion. The energy is then passed down to smaller-sized eddies until it eventually reaches the molecular scale and is dissipated as heat (i.e. it

returns to the thermal portion of the solar energy cycle). This energy does not appear in Figure 1.8 because on an annual basis there is a balance between kinetic energy production and dissipation.

In the boundary layer we are concerned both with motion generated on the micro- and local scales, and with the modification of existing airflow generated on scales larger than those of the boundary layer. In the first category we are concerned with wind systems generated by horizontal thermal differences in the boundary layer. These local scale thermal winds are especially prevalent across the boundaries between contrasting surface types. Examples include the breezes occurring at land/sea (lake), mountain/valley, forest/grassland, and urban/rural interfaces. In the second category we are concerned with the role of surface roughness in shaping the variation of wind speed with height, and with the way in which uneven terrain (e.g. hills and valleys) and isolated obstacles (e.g. a tree or a building) perturb existing flow patterns. All of these aspects are considered in Chapter 5.

## 4 MASS BALANCES

### (a) Properties of water

Water possesses a number of unusual properties which make it an important climatological substance. One important thermal property is its high *heat capacity* (see p. 36 and Table 2.1, p. 44). This effectively means that in comparison with most other natural materials it takes much more energy input to cause a similar rise in the temperature of water. Equally, subtraction of energy does not cause water to cool as rapidly. This property makes water a good energy storer, and a conservative thermal influence.

Water is the only substance that exists in all of its states at temperatures normally encountered in the E-A system. In changing between ice, water and water vapour, latent heat is taken up or liberated and as a result the energy and water balances become enmeshed. The energy required to effect a change between the ice and water phases (i.e. consequent upon melting or freezing) is  $0.334 \text{ MJ kg}^{-1}$  at  $0^\circ\text{C}$ , and is called the *latent heat of fusion* ( $L_f$ ). The change between liquid water and water vapour (i.e. consequent upon evaporation or condensation) at  $0^\circ\text{C}$  requires  $2.50 \text{ MJ kg}^{-1}$  which is almost 7.5 times more energy. This is the value of the latent heat of vaporization ( $L_v$ ), and at  $10^\circ\text{C}$  it is  $2.48$ , at  $20^\circ\text{C}$   $2.45$  and at  $30^\circ\text{C}$   $2.43 \text{ MJ kg}^{-1}$  (see Appendix A3, p. 393). In the event that the water changes directly between the ice and vapour phases (i.e. sublimates) the *latent heat of sublimation* ( $L_s$ ) is the algebraic sum of  $L_f$  and  $L_v$ , and at  $0^\circ\text{C}$  it is  $2.83 \text{ MJ kg}^{-1}$ . To gain some measure of the energy amounts involved it should be realized that the energy locked-up in evaporating 1 kg of water is roughly equivalent to that necessary to raise 6 kg of water from  $0^\circ\text{C}$  to  $100^\circ\text{C}$ .

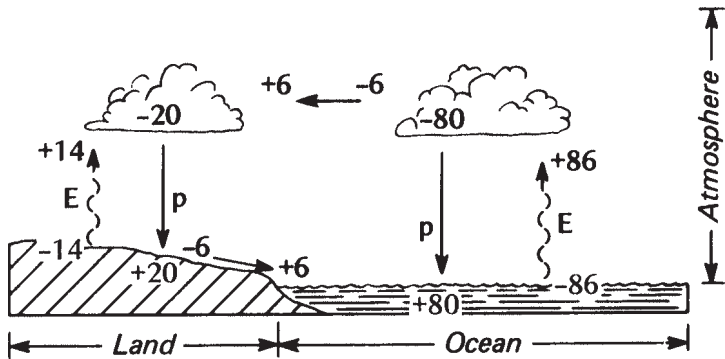


Figure 1.13 Schematic diagram of the average annual hydrologic cycle of the Earth-Atmosphere system. Values expressed as percentages of the mean annual global precipitation of 1040 mm (data from Chow, 1975).

### (b) Water balance

The annual global cycle of water in the E-A system is given in Figure 1.13. All quantities of water are represented as percentages of the mean annual global precipitation, which is approximately 1040 mm. Utilizing energy provided by the energy balance (Figure 1.8) water is evaporated from open water surfaces (oceans and lakes) and the soil, and is transpired from vegetation. The composite loss of water to the air from all sources is called the *evapotranspiration* ( $E$ ). The water vapour is carried up into the Atmosphere by unstable air masses, and mechanical convection. Eventually the vapour is cooled to its dew-point (p. 64), and it condenses as a cloud droplet, or ice crystal. Under favourable conditions the cloud droplets or crystals may grow to a size where they can no longer be held in suspension and they fall to the Earth as *precipitation* ( $p$ ). Near the surface water may also be deposited by direct condensation or sublimation as dew, hoar frost, and rime, or be impacted as fog-drip. Over land areas  $p$  is greater than  $E$ , and the excess is transported as streamflow to the oceans where  $E$  is greater than  $p$  (Figure 1.13). So that for the Land and Ocean sub-systems their annual water balance may be written:

$$p = E + \Delta r$$

where,  $\Delta r$ —*net runoff* (i.e. the net change in runoff over a distance). This term may have a positive or negative sign. It is positive if more water leaves than arrives as is normally the case on sloping land.  $\Delta r$  is negative when surface flow leads to accumulation of water such as when lake levels rise. For the total E-A system on an annual basis the balance is even simpler:

$$p = E$$

because the system is closed to the import or export of mass and hence all horizontal transfers (such as runoff or ocean currents) are internal, and the net storage change for the system is zero.

We will accept the macro-scale conditions as given, and concentrate on small-scale surface/atmosphere interaction over relatively short time periods. Let us return to the case of our 'ideal', or short grass, site with a moist soil on level terrain. If we consider the water exchanges through the surface plane (Figure 1.14a) then we can formulate the surface water balance equation as:

$$p = E + f + \Delta r \tag{1.18}$$

where  $f$  -infiltration to deeper soil layers. Normally  $f$  is positive due to gravity. Although less common  $f$  can be negative, for example, where a ground water table intersects a hillslope leading to spring seepage. Infiltration is not easily determined so for practical purposes it is better to consider a column (Figure 1.14b) which extends from the surface to a depth where

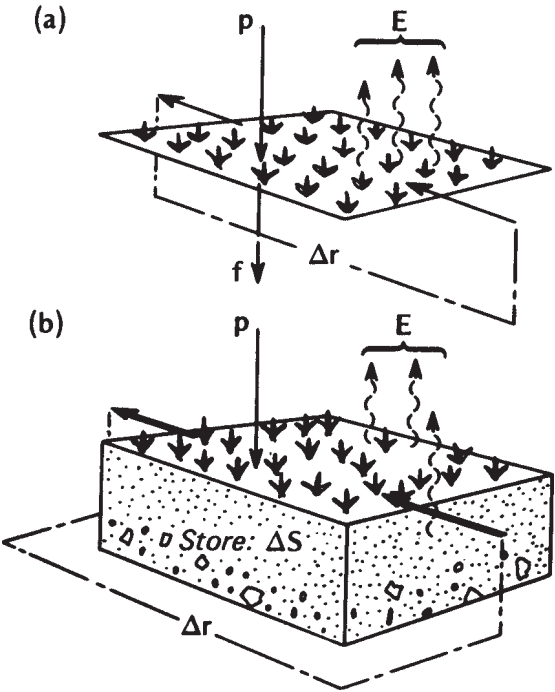


Figure 1.14 Diagrammatic representation of the components of the water balance of (a) a natural surface, and (b) a soil-plant column.

significant vertical exchanges are absent (i.e. where  $f \rightarrow 0$ ), then the water balance is given by:

$$p = E + \Delta r + \Delta S \quad (1.19)$$

where  $\Delta S$ —the net change in *soil moisture content*. The soil moisture content is a measure of the mass of water stored in a soil in the same way as soil temperature is a measure of the soil heat content. Equation 1.19 shows how the water storage in the system is dependent upon the water input which is usually mainly  $p$ , and the water output via  $E$  and  $\Delta r$ . Input could also be supplied by irrigation which would require an additional term on the left-hand side of equation 1.19, or it could be as dewfall (i.e. convective transfer from the air to the surface— $E$ ). Evapotranspiration consists of evaporation of free surface water (e.g. puddles), and soil pore water, and water transpired from vegetation.

The time scale over which equation 1.19 is valid proves awkward if we are properly to integrate it with the surface energy balance (equation 1.17) on time periods of a day or less. This arises because the input/output processes are fundamentally different in nature. Precipitation usually occurs in discrete, short-period bursts, whereas evaporation is a continuous and variable function. Thus, for example, during periods with no precipitation water input is zero but the soil moisture store is being almost continually depleted by evapotranspiration. In these circumstances equation 1.19 effectively reduces to:

$$E = \Delta S \quad (1.20)$$

because  $\Delta r$  is negligible on level terrain. Therefore, unlike the annual situation where net water storage is zero, on the short time-scale  $\Delta S$  is non-zero and very important.

Soil moisture is significant in surface energy balance considerations because it is capable of affecting radiative, conductive and convective partitioning. For example, the addition of moisture can alter the surface albedo thereby changing  $K^*$  and  $Q^*$ . Equally the thermal properties of a soil are changed by adding water, so that heat transfer and storage are affected. Most important however are the potential latent heat effects of soil moisture.

The common term in the water and energy balance equations is evaporation. The fluxes of mass ( $E$ ) and energy ( $Q_E$ ) associated with evaporation are linked by the relation:

$$Q_E = L_v E \quad (1.21)$$

where the units of  $E$  are  $\text{kgm}^{-2}\text{s}^{-1}$  (mass transport through unit surface area in unit time). Therefore if the energy scale of Figure 1.10 were divided by  $L_v$  the curve of  $Q_E$  becomes the diurnal course of  $E$ .



## 32 Boundary Layer Climates

When temperatures are at or below 0°C and the change of water phase is to or from the solid state (ice), the equivalent expression is:

$$\Delta Q_M = L_f M \quad (1.22)$$

Here  $\Delta Q_M$  is the energy flux density needed to melt water at the mass flux density  $M$  ( $\text{kgm}^{-2}\text{s}^{-1}$ ), or it is the energy flux density that is released upon freezing.

For hydrologic purposes it is convenient to express the mass flux densities  $E$  and  $M$  in terms of an equivalent depth of water (millimetres) over the period of concern (usually an hour or a day). They are then consistent with the normal units of precipitation. Details of the conversion factors involved are given in Appendix A4e (p. 398). The simple conversion  $1 \text{ mm evaporation} = 2.45 \text{ MJm}^{-2}$  has been applied to the daily total value of  $Q_E$  in the table accompanying Figure 1.10 to yield the daily water loss in millimetres.

It is also satisfying to note that conversion of the  $Q_E$  annual energy term in Figure 1.8 to an equivalent height of water evaporated gives a value of  $E$  very close to that in Figure 1.13. Thus loss of water to the air not only depletes the mass store (soil moisture) but also the energy store (soil and air temperature) as a result of taking up latent heat. Condensation operates in the reverse sense by adding to both the mass and energy stores. Melting and freezing are energetically less significant, but still of importance especially in soil climate.

The measures of soil moisture content, and the processes of soil moisture movement and evaporation, are outlined in Chapter 2.

### (c) Other mass balances

The cycles of energy and water are by far the most important in explaining surface climates, but it should be noted that there are others operating in the E-A system on similar space and time scales. Examples include the cycles of carbon, nitrogen, oxygen and sulphur. Of these the  $\text{CO}_2$  portion of the carbon cycle is of most immediate interest to this book because it interacts with both solar energy and water in the process of photosynthesis. This is dealt with in Chapter 4 and an example of the diurnal flux of  $\text{CO}_2$  is given in Figure 4.10.

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